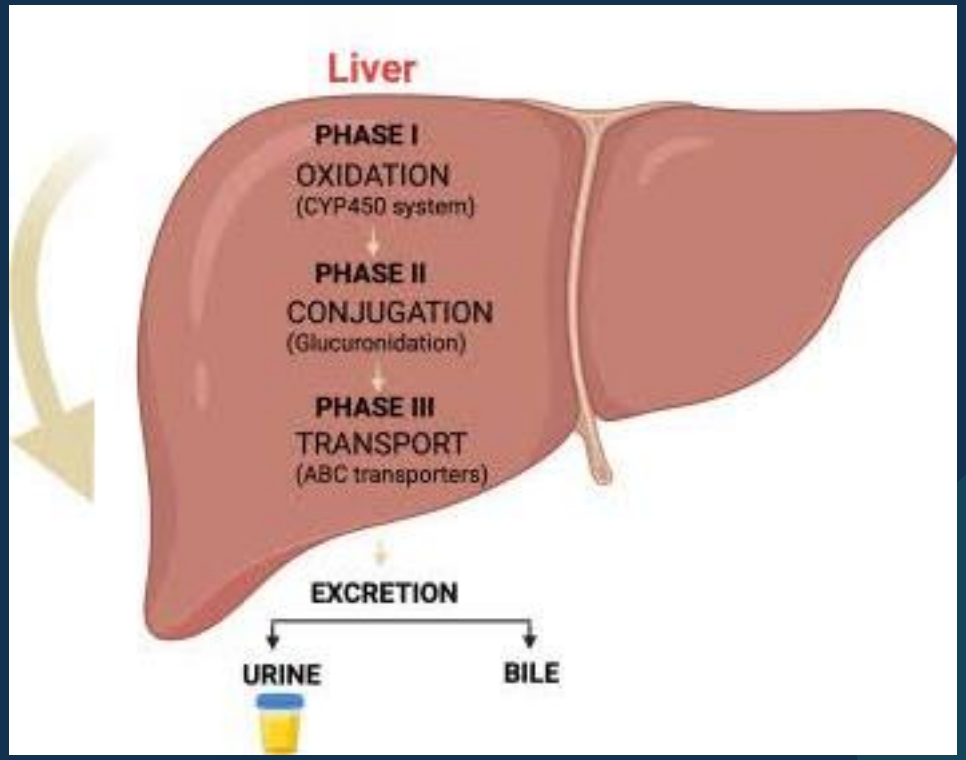
PHARMACOKINETICS — CONTINUED

Drug Metabolism

(Biotransformation)

Nermin Jamal

MSc. Pharmacology



LECTURE ROADMAP

Objectives

1. Why drug metabolism occurs
2. What are the results of drug metabolism
3. What are the enzymes responsible for drug metabolism
4. How drug metabolism occurs
5. What are the drugs that induce or inhibit these enzymes

FOUNDATIONS

What Is Drug Metabolism?

Metabolism and excretion are processes responsible for eliminating drugs from the body. The three major routes of elimination are **hepatic metabolism, biliary elimination, and urinary elimination.**

Drug Metabolism (Biotransformation)

Also called biotransformation, is one of the primary mechanisms by which drugs are inactivated and become more polar, hydrophilic, and water soluble in order to be ready for excretion by the kidneys.

The liver is the major site for drug metabolism, but specific drugs may undergo biotransformation in other tissues, such as intestine, lung, skin and kidney.

INTESTINAL METABOLISM

Clonazepam and chlorpromazine are more extensively metabolized by the intestine.

GASTRIC METABOLISM

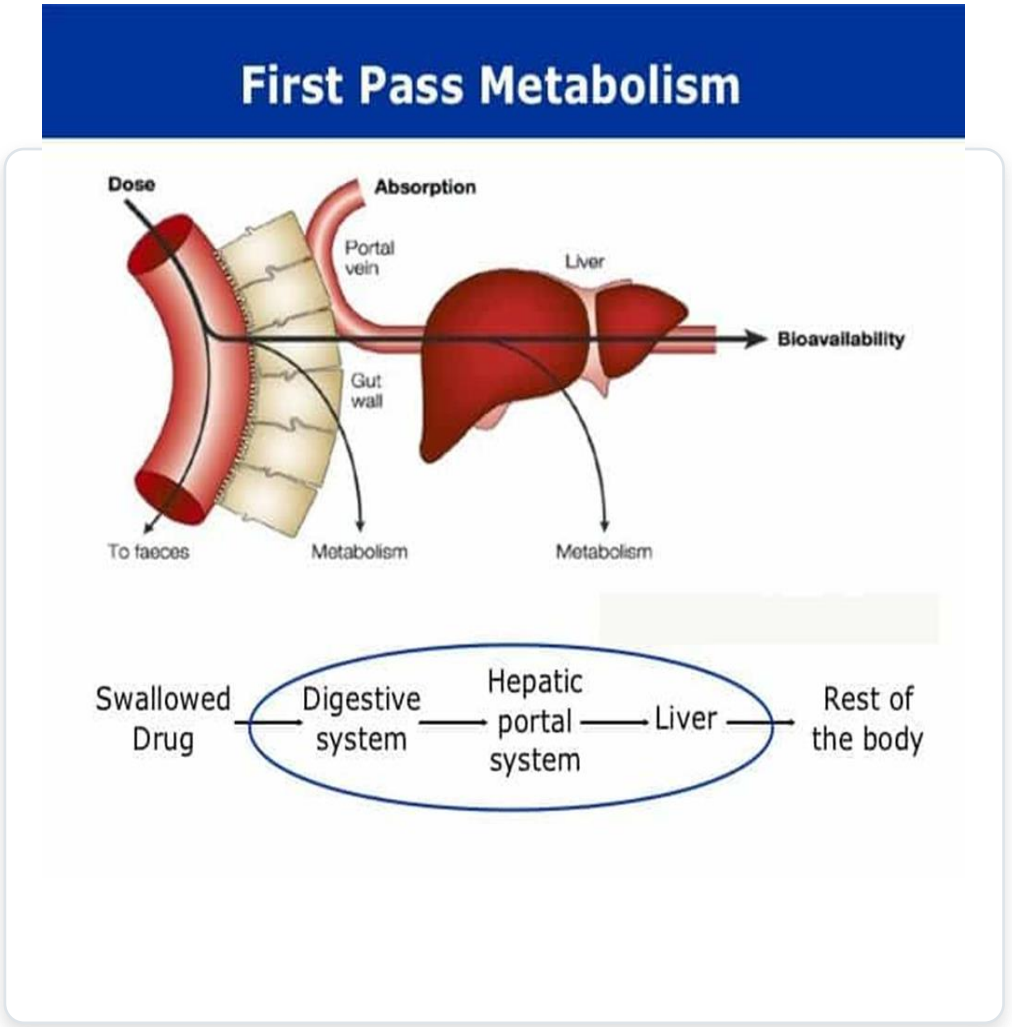
Penicillin is metabolized by gastric acid.

First-Pass Metabolism

Biotransformation of a drug by intestinal or liver enzymes before they reach systemic circulation, **resulting in lower systemic bioavailability of the parent drug and diminished therapeutic effect.**

BYPASSING FIRST-PASS EFFECT

First-pass effect may be bypassed if the drug is administered IV or sublingually, e.g., nitroglycerin, morphine, and Fentanyl — which is never given orally.



Consequences of Drug Metabolism

A Active → Inactive

Conversion of a pharmacologically active drug to an inactive product — most drugs are inactivated by metabolism.

B Inactive (Prodrug) → Active

Methanol → formaldehyde
Prednisone → prednisolone
Sulfasalazine → 5-aminosalicylic acid

C Active → Another Active Metabolite

Codeine → morphine
Digitoxine → digoxine

D Non-toxic → Toxic Metabolite

Paracetamol is converted into the toxic metabolite N-acetyl-p-benzoquinone imine (NAPQI).

Rates of Metabolism

In the liver, drugs are metabolized at various rates, either by first-order kinetics or zero-order kinetics.

FIRST-ORDER KINETICS

- A **constant fraction** of the drug is metabolized per unit of time.
- The rate of metabolism increases proportionately as the concentration of the drug in the body increases.
- The enzymes involved in metabolizing most drugs are not saturable at normal drug concentrations.
- Most drugs are metabolized by first-order kinetics.

ZERO-ORDER KINETICS

- **A fixed amount** of drug can be metabolized at any one time because the enzymes involved are saturable.
- Only a few drugs (e.g., alcohol and phenytoin) follow zero-order kinetics for metabolism.
- The rate of metabolism is constant and does not vary with the amount of drug present.

ENZYME SYSTEMS

Drug Metabolizing Enzymes

MICROSOMAL ENZYMES

Like cytochrome P450 enzymes, located at the smooth endoplasmic reticulum primarily of the liver and also in the kidney, intestinal mucosa, and lungs.

They catalyze:

- Oxidation, reduction, hydrolysis (Phase I reactions)
- Glucuronosyltransferase (the only inducible Phase II reaction) is also located inside liver microsomes

Inducible enzymes

NON-MICROSOMAL ENZYMES

Like Monoamine oxidases (MAO), Esterases, Amidases, Transferases, Conjugases — located in the cytoplasm and mitochondria of hepatic cells as well as in other tissues including plasma.

They catalyze:

- All Phase II reactions except glucuronidation

Non-inducible enzymes

Phase I Reactions

- Convert the parent compound into a more polar (hydrophilic) metabolite by adding or unmasking functional groups (-OH, -SH, -NH2, -COOH, etc.)
- Usually inactivates the parent compound
- If sufficiently polar, the metabolite is excreted
- If not sufficiently polar, it undergoes a Phase II reaction

P450 SYSTEM

a — Phase I reactions utilizing the P450 system

The Phase I reactions most frequently involved in drug metabolism are catalyzed by the cytochrome P450 system (also called microsomal mixed-function oxidase). Their activity requires both a reducing agent (NADPH) and molecular oxygen.

$$\text{Drug} + O_2 + \text{NADPH} + H^+ \rightarrow \text{Drug-OH} + H_2O + \text{NADP}^+$$

b — Phase I Reactions Not Involving P450

These include amine oxidation by the MAO enzyme (e.g., oxidation of catecholamines or histamine), alcohol dehydrogenation (e.g., ethanol oxidation), esterases (e.g., metabolism of pravastatin in the liver), and hydrolysis (e.g. of procaine).

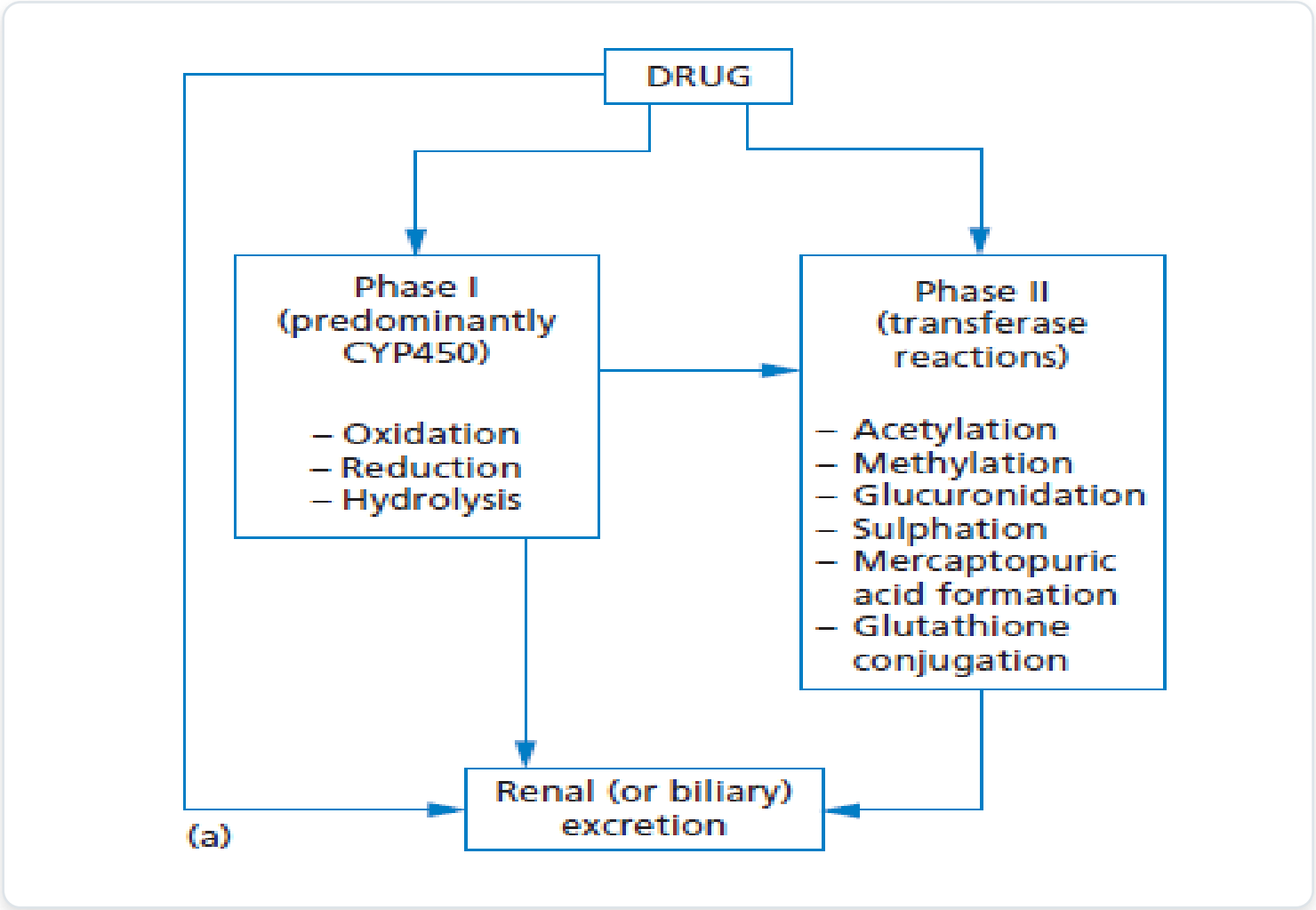
Phase II Reactions

Also called biosynthetic reactions — an endogenous substrate is conjugated (combined) with the Phase I metabolite to further increase aqueous solubility, e.g., conjugation with glucuronide, sulfate, acetate, or amino acid.

The enzymes are called transferases, e.g., Glucuronyl transferase, Sulfotransferase, N-Acetyltransferase, etc.

Glucuronyl transferase is responsible for the majority of Phase II reactions.

Phase I & Phase II Pathway



What Phase II Metabolism Produces

- A highly polar conjugate
- Generally inactive — with an exception to the rule: the morphine glucuronide metabolite is a more potent analgesic than the parent compound
- Rapidly excreted in the urine

ENTEROHEPATIC RECIRCULATION

High molecular weight conjugates excreted in the bile may have their conjugate bond cleaved by intestinal flora, releasing the parent drug back into the systemic circulation; this process is called “enterohepatic recirculation”

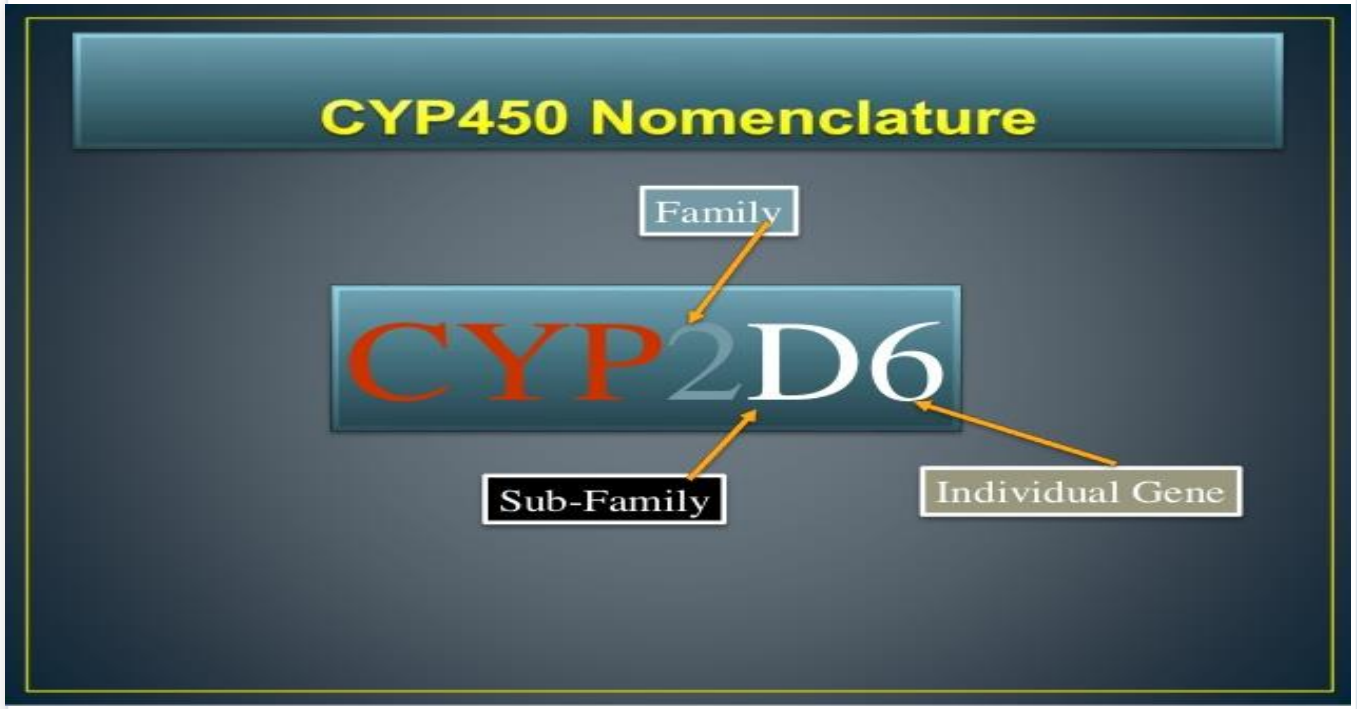
- **Delayed parent drug elimination**
- **Prolongation of drug effect**

ORDER CAN VARY

Phase I usually precedes Phase II, but not all drugs undergo the two phases in that order.

Example: isoniazid is first acetylated (N-acetylation, a Phase II reaction) and then hydrolyzed to isonicotinic acid (a Phase I reaction).

CYP450 Nomenclature



Family, sub-family, and individual gene designations within a CYP isoform name.

The Cytochrome P450 System

- The P450 system is a superfamily of hemoproteins, important for the metabolism of many endogenous compounds (steroids, lipids, etc.) and for the biotransformation of exogenous substances (xenobiotics). They are located in most cells but are primarily found in the liver and GI tract.
- 3 main CYP families in the human liver: CYP1, CYP2, CYP3
- Six isozymes are responsible for the vast majority of P450-catalyzed reactions: CYP3A4, CYP2D6, CYP2C9/10, CYP2C19, CYP2E1, and CYP1A2

CYP	% Total Human Liver P450 Content
1A2	15
2B6	1
2C9	20
2D6	**5**
2E1	10
3A4	**30**

CYP450 ENZYMES

Major CYP Isoforms & Substrates

~50%

CYP3A4

Most predominant isoform, accounting for about 50% of all drug oxidations. Considerable amounts are found in intestinal mucosa, accounting for first-pass metabolism of drugs such as chlorpromazine and clonazepam.

~30%

CYP2D6

Accounts for about 30% of drug oxidations. Substrates include some opioid analgesics (e.g., codeine) and some antidepressants (e.g., fluoxetine).

~20%

CYP2C9

Accounts for about 20% of oxidation. Substrates include phenytoin & warfarin (narrow therapeutic indices) and NSAIDs (e.g. ibuprofen).

Other

Remaining Isoforms

Remaining CYP isoforms metabolize fewer drugs. CYP2E1 metabolizes ethanol and acetaminophen.

MODULATING FACTORS

Factors Affecting Drug Metabolism

1. **Modulation of Liver Microsomal Enzymes**
   Enzyme Inhibition • Enzyme Induction
2. **Disease**
3. **Age and Gender**
4. **Genetic Variation**

Enzyme Induction

Cytochrome P450 Enzyme Induction: Some drugs or substances, following repeated administration, increase the activity of microsomal oxidase systems (increase the amount of P450 enzymes). Induction takes time to develop.

EXAMPLE INDUCERS

- Alcohol
- Cigarette smoking
- Barbiturates (hypnotic phenobarbitones)
- Phenytoin (antiepileptic; most of them are inducers)
- Rifampicin
- Clotrimazole, Griseofulvin (antifungal)
- Spironolactone

HOW INDUCTION OCCURS

Increased Enzyme Synthesis Rate

E.g. phenobarbital, rifampin, and carbamazepine are capable of increasing the synthesis of one or more CYP isozymes.

Reduced Enzyme Degradation Rate

E.g. clotrimazole-mediated induction of CYP3A4 enzymes.

Consequences of Induction

1. **Increase the metabolism of the inducer.**

2. **Tolerance**

   Decreased plasma levels of the parent compound decrease the inducer's pharmacological (therapeutic) effect.

3. **Increase the metabolism of co-administered drugs**

   Drug interaction — e.g. barbiturates and warfarin (thrombosis); phenytoin and oral contraceptives (the woman gets pregnant).

4. **As therapy**

   E.g., Phenobarbitone is given to babies with physiological jaundice/hyperbilirubinemia to induce liver microsomal enzymes.

Enzyme Inhibition

Inhibition of CYP isozyme activity is an important source of drug interactions that lead to serious adverse events. Enzyme inhibition can occur after a short period of exposure to the inhibiting drug. The most common form of inhibition is through competition for the same isozyme.

EXAMPLE INHIBITORS

- Omeprazole, Cimetidine (anti-peptic ulcer)
- Erythromycin, ciprofloxacin (macrolides)
- Ketoconazole (antifungal)
- Grapefruit
- Probenecid

CLINICAL IMPACT

Omeprazole is a potent inhibitor of the CYP isozymes responsible for warfarin metabolism. Taken together, plasma warfarin concentrations rise, increasing the risk of hemorrhage and other serious bleeding reactions.

Cimetidine blocks the metabolism of theophylline, clozapine, and warfarin.

Natural substances such as grapefruit juice may also inhibit drug metabolism.

Consequences of Inhibition

CONSEQUENCES OF INHIBITION

- Increased plasma levels of the parent compound
- Prolonged pharmacological effects of the parent compound
- Enhanced drug-induced toxicity

Factors Affecting Drug Metabolism (cont….)

2. DISEASE

Liver disease (e.g. hepatitis, liver cirrhosis, altered liver perfusion secondary to cardiac disease) mainly affects metabolism because the liver is the major site of drug metabolism — severity of disease determines the extent of reduced metabolism.

Severe cirrhosis: 30–50% reduction in metabolism

Bioavailability of drugs that undergo extensive first-pass metabolism may increase 3–4 fold with disease.

3. Age and Gender

- Phase I & II enzymes require maturation it is functional in the fetus, but not mature; full function matures slowly after birth.
- As aging occurs, there is a slow decline in drug metabolism.
- **Age affects Phase I metabolism, but not Phase II.**
- Decline in the elderly may be due to additional factors, e.g. liver disease, heart failure.

GENDER & PREGNANCY

The influence of gender on drug metabolism is minor compared to other factors. Ethanol undergoes more extensive first-pass metabolism in men than in women.

During the second and third trimesters of pregnancy, induction of metabolism occurs, and drug doses may need to be increased for certain drugs.

MODULATING FACTORS

4. Genetic Variation of Metabolism

Genetic diversity occurs in the enzymes that control the metabolism of drugs and other chemicals.

Complete Lack of Activity

Reduced Activity

Gene Duplication & Increased Activity

METABOLIZER PHENOTYPES

- **PM** — Poor Metabolizers
- **EM** — Extensive Metabolizers
- **UM** — Ultrarapid Metabolizers

SUMMARY

To terminate the effect of lipophilic drugs which are not excreted by kidney, they are metabolized through phase I and phase II reactions that are catalyzed by liver enzymes, especially CYP450, and these enzymes may be induced or inhibited by drugs.